

Artificial Intelligence (Spring 2024)

AI Week 10

Dataset Preprocessing Techniques

[Mean Imputation, Z-Score, Normalization]



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Mean Imputation

Mean Imputation:

A technique used in statistics to **fill in missing values** in a **data set**. This is done by **replacing the missing value** with the **mean of the remaining values** in the data set.

Advantages of Mean Imputation

- **Replacement of missing data** with a reasonable value.
- Help to **reduce the bias** in the results.



Disadvantages of mean imputation

- It can lead to **inaccurate estimates** of variability and standard errors.
- It can produce imputations that are **not representative** of the underlying data.



Mean Imputation in Python

```
import pandas as pd
import numpy as np
from sklearn.impute import SimpleImputer

df = pd.DataFrame()

# add two features with 1 missing value each
df['f1'] = [1,4,2,3,4,5,6,np.nan,5,4]
df['f2'] = [np.nan,2,2,5,4,8,5,3,6,5]

print(df)
```

	f1	f2
0	1.0	NaN
1	4.0	2.0
2	2.0	2.0
3	3.0	5.0
4	4.0	4.0
5	5.0	8.0
6	6.0	5.0
7	NaN	3.0
8	5.0	6.0
9	4.0	5.0

Mean Imputation in Python

```
# Initialize the imputers, by setting what values we want to impute and the strategy to use  
mean_imputer = SimpleImputer(missing_values=np.nan, strategy='mean')
```

```
# Fit the imputer on to the dataset  
mean_imputer = mean_imputer.fit(df)
```

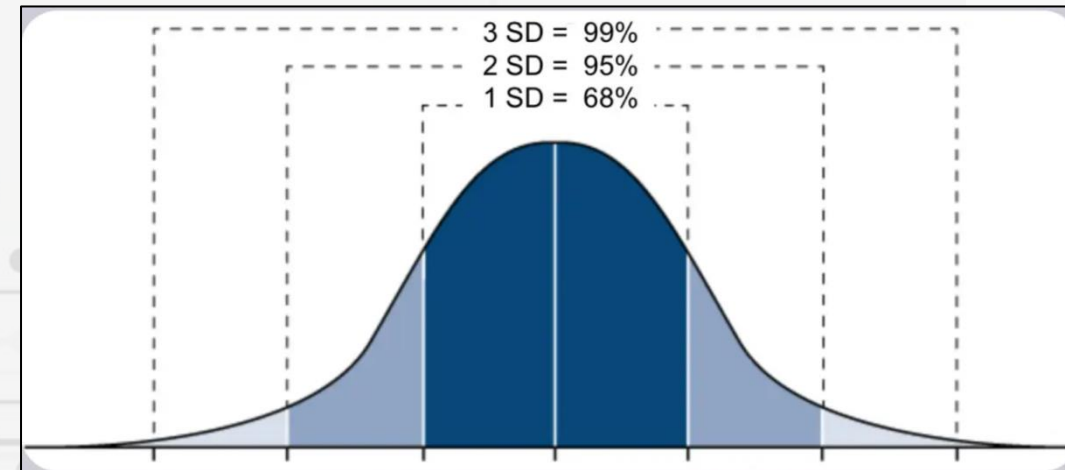
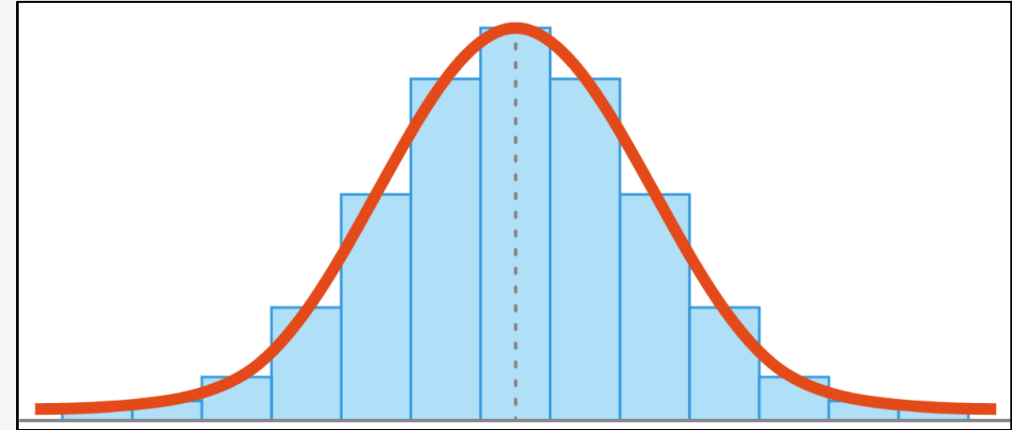
```
# Apply the imputation  
results = mean_imputer.transform(df.values)  
results.round()
```

```
array([[1., 4.],  
       [4., 2.],  
       [2., 2.],  
       [3., 5.],  
       [4., 4.],  
       [5., 8.],  
       [6., 5.],  
       [4., 3.],  
       [5., 6.],  
       [4., 5.]])
```

Z-Score

Z score:

- Z score is also called **standard score**.
- This score helps to understand if a data value is greater or smaller than mean and **how far away it is from the mean**.
- More specifically, Z score tells **how many standard deviations away a data point is from the mean**.



Z score and Outliers

- If the **z score** of a data point is **more than 3**, it indicates that the data point is quite different from the other data points. Such a **data point can be an outlier**.
- For example, in a survey, it was asked how many children a person had. Suppose the data obtained from people is

1, 2, 2, 2, 3, 1, 1, 15, 2, 2, 2, 3, 1, 1, 2

Clearly, 15 is an outlier in this dataset.

Z Score in Python

```
import numpy as np
```

```
data = [1, 2, 2, 2, 3, 1, 1, 15, 2, 2, 2, 3, 1, 1, 2]  
mean = np.mean(data)  
std = np.std(data)  
print('mean of the dataset is', mean)  
print('std. deviation is', std)
```

Output:

```
mean of the dataset is 2.6666666666666665  
std. deviation is 3.3598941782277745
```

```
threshold = 3  
outlier = []  
for i in data:  
    z = (i-mean)/std  
    if z > threshold:  
        outlier.append(i)  
print('outlier in dataset is', outlier)
```

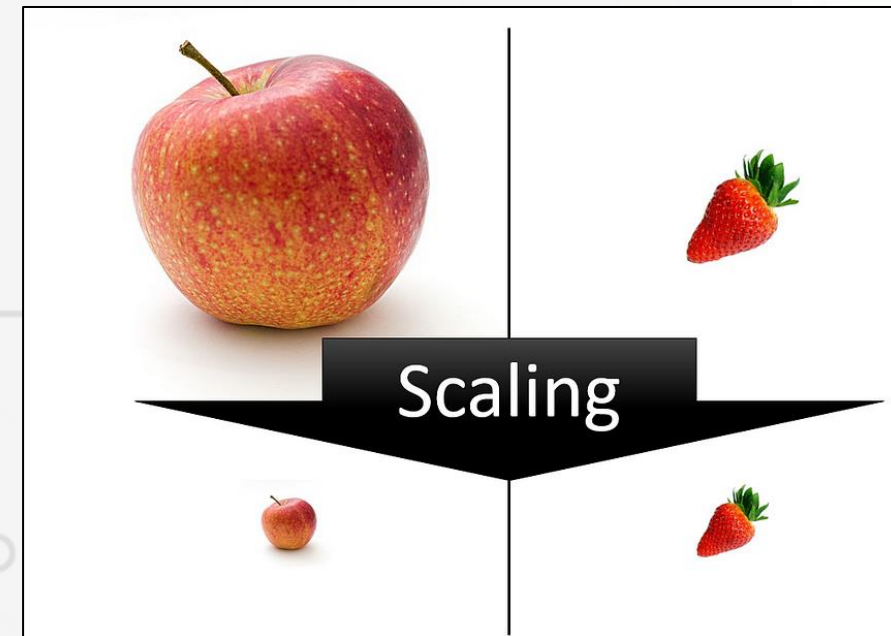
Output:

```
outlier in dataset is [15]
```

Data Normalization

Data normalization:

- A technique used in **data processing** that helps to **transform and organize data into a structured format**.
- The process involves scaling the values of **different features** or variables in a dataset to be on a **similar scale**.
- This ensures that **no particular feature dominates** or influences the analysis more than others, **leading to unbiased and accurate results**.



Data Normalization

- Normalization is particularly useful when working with **datasets that have features with significantly different ranges**, as it can help **bring them to the same level of importance**.
- It is also beneficial when using **machine learning algorithms** that are **sensitive to the scale** of the input data, such as **k-nearest neighbors** and **neural networks**.



Different Scales Example

- For instance, imagine a dataset with two features: **age** and **income**.

Age is typically measured in **years**, while **income** is measured in **dollars**.

- The values for age would be much smaller than the values for income, which can range from a few thousand dollars to millions.

Customer	Age	Income	No. credit cards	Class
George	35	35K	3	No
Rachel	22	50K	2	Yes
Steve	63	200K	1	No
Tom	59	170K	1	No
Anne	25	40K	4	Yes
John	37	50K	2	YES

Normalization in Python

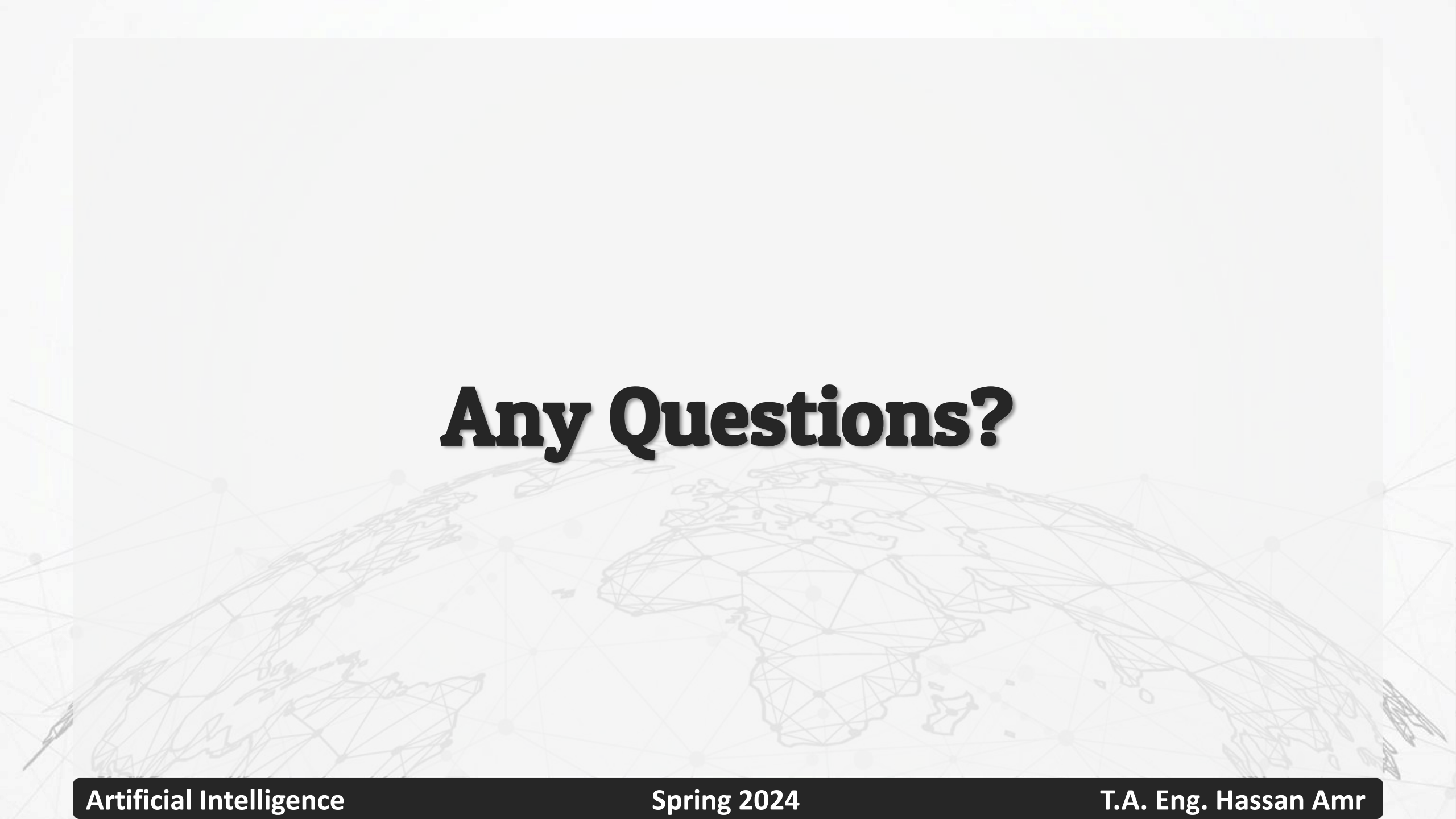
```
from sklearn.preprocessing import StandardScaler
import pandas as pd

# Load dataset
data = pd.read_csv('dataset.csv')

# Extract numerical features
num_features = ['feature1', 'feature2', 'feature3']
X = data[num_features]

# Normalize data using StandardScaler
scaler = StandardScaler()
X_normalized = scaler.fit_transform(X)

# Print normalized data
print(X_normalized)
```

The background of the slide features a stylized, light gray globe. Overlaid on the globe is a complex network of thin gray lines connecting various points, resembling a global communication or data network. The text "Any Questions?" is centered in a large, bold, black font with a slight drop shadow.

Any Questions?